



Unusual uses of Ansible Automation Platform and Event-Driven Ansible



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Who're we?

- Atea Denmark has a department called Open Source
- We work with... open source!
 - Linux (all flavors)
 - PostgreSQL, MariaDB, Redis
 - Kubernetes, Docker, Podman
 - Ansible, Puppet
 - DevOps, GitOps, et al.

Who am I?

- Rune Juhl
- Architect, Open Source, Atea Denmark
- Linux user for > 22 years
- Rabid advocate of open-source software
- Emacs user



What is this about?

- Playing around with Ansible and EDA because
 - It's fun!
 - It's educational!
 - ...because why not?
- Beware!
 - This is exploration, not guaranteed to work
 - Not polished
 - No AI

Why home automation?

- Been using Home Assistant for a long time
- "Programming" in YAML is not fun
- CLI >> GUI
- ...because I felt like it!



Link to "Why the fuck are we templating yam!?", a blog post by Lee Briggs

<https://leebriggs.co.uk/blog/2019/02/07/why-are-we-templating-yaml>

GUI fatigue

Blueprint

Blueprint to use

Controller - IKEA E1743 TRÅDFRI On/Off Switch & Dimmer

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Controller automation for executing any kind of action triggered by the provided IKEA E1743 TRÅDFRI On/Off Switch & Dimmer. Allows to optionally loop an action on a button long press.

Supports deCONZ, ZHA, Zigbee2MQTT.

Automations created with this blueprint can be connected with one or more [Hooks](#) supported by this controller.

Hooks allow to easily create controller-based automations for interacting with media players, lights, covers and more.

See the list of [Hooks available for this controller](#) for additional details.

Full documentation regarding this blueprint is available [here](#).

This blueprint is part of the [Awesome HA Blueprints](#) project.

Version 2023.10.05

(Required) Integration

Integration used for connecting the remote with Home Assistant. Select one of the available values.

deCONZ

ZHA

Zigbee2MQTT

(deCONZ, ZHA) Controller Device

The controller device to use for the automation. Choose a value only if the remote is integrated with deCONZ, ZHA.

Device

(Zigbee2MQTT) Controller Entity

The action sensor of the controller to use for the automation. Choose a value only if the remote is integrated with Zigbee2MQTT.

Entity

ikea_e1743_switch_01 Action

(Required) Helper - Last Controller Event

Input Text used to store the last event fired by the controller. You will need to manually create a text input entity for this, please read the blueprint Additional Notes for more info.

Entity

ikea_e1743_switch_01_helper

(Optional) Up button short press

Action to run on short up button press.

Perform action 'Light: Turn on' on Bedroom ceiling lamp

+ ADD ACTION

+ ADD BUILDING BLOCK

(Optional) Up button long press

Action to run on long up button press.

+ ADD ACTION

+ ADD BUILDING BLOCK

(Optional) Up button release

Action to run on up button release after long press.

+ ADD ACTION

+ ADD BUILDING BLOCK

(Optional) Up button double press

Action to run on double up button press.

Light 'Turn on' on Bedroom

+ ADD ACTION

+ ADD BUILDING BLOCK

YAML fatigue

```
174 condition:
175   # if triggered by the homeassistant platform check if the automation should be run, as specified by the user
176   # if triggered by something else, just continue
177   - condition: and
178     conditions:
179       # if triggered by the run time, check that the automation is allowed to run today
180       - >-
181         {%- set today = now().weekday() -%} {{ trigger.id != "run_time" or today == 0 and day_monday or today == 1 and day_tuesday or
182         # if triggered due to home assistant startup, check that the feature is enabled
183         - '{{ trigger.id != "homeassistant_start" or trigger_at_homeassistant_startup }}'
184         # if triggered due to home assistant startup or to a custom trigger event, perform the execution check
185         # check whether the automation run at or after the expected run time
186         # if this is the case, there's no need to run the automation again
187         # else, the automation was not able to run at the expected time, and was not executed after it, hence run it now
188         # if block late execution is enabled, check also that no more than the provided max elapsed time has passed since the expected r
189         - >-
190         {%- set cdt = now() | as_timestamp | timestamp_local | as_datetime -%}
191         {%- set lsrdt = ((states(helper_storage) | from_json).last_triggered if (states(helper_storage) | regex_match(json_regex)) els
192         {%- set idt = strptime(states(time_entity),time_fmt) -%}
193         {%- set days = [day_monday, day_tuesday, day_wednesday, day_thursday, day_friday, day_saturday, day_sunday] -%}
194         {%- set cwd = cdt.weekday() -%}
195
196         {% macro day_offset(day) -%}
197           {%- if days[day] -%}
198             {{ 7 if cwd == day and cdt.time() < idt.time() else (cwd-day)%7 }}
199           {%- else -%}
200             -1
201           {%- endif -%}
202         {%- endmacro %}
203
204         {%- set weekdays = [day_offset(0)|int, day_offset(1)|int, day_offset(2)|int, day_offset(3)|int, day_offset(4)|int, day_offset(
205
206         {%- if weekdays | length > 0 -%}
207           {%- set closest_weekday = weekdays | min -%}
208           {%- set day_diff = timedelta(weeks=weeks_frequency-1, days=closest_weekday) -%}
209           {%- set edt = (cdt - day_diff - timedelta(hours=cdt.hour, minutes=cdt.minute, seconds = cdt.second) + timedelta(hours=idt.ho
210             {{ trigger.id not in ["homeassistant_start","custom_trigger_event"] or lsrdt < edt and (not block_late_execution or edt >= c
211           {%- else -%}
212             {{ false }}
213           {%- endif -%}
214
215 action:
216   # store the current datetime into the helper, since the automation was successfully triggered
217   - service: input_text.set_value
218     data:
219       entity_id: !input helper_storage
```

A large orange circle is positioned on the left side of the slide, partially cut off by the edge.

Let's replace
it!

- ...with Ansible and more YAML!
 - ...maybe more flexible?
 - No restrictions on programming languages
 - Lots of Ansible modules
 - How hard can it be?
 - ...he said, before spending countless hours debugging and monkey-patching EDA Decision Environments
- 
- A blue dashed line is located in the bottom right corner of the slide, consisting of several short, curved segments.



Home automation with Ansible and EDA!

Trigger sources

- MQTT
 - Zigbee2MQTT
 - DNS server logs
- HTTP server-sent events (SSE)
 - ESPHome devices
- Websocket
 - Home Assistant
- Polling HTTP

Respond using Ansible

- MQTT
 - Home Assistant
- HTTP REST API
 - UniFi Network Application (undocumented?)
 - Home Assistant (slightly less undocumented)
 - Drinks machine
- Anything you can think of

Ideas



Listen to presence sensor,
turn on lights



Throttle Wi-Fi if the kids play
Roblox at odd hours



Detect loo roll empty (TP
LOAD LETTER)



Hook up drinks machine and
have it dispense drinks when
you get home



Ingest events from mm-Wave
radar, turning on smoke
machine when Halloweeners
nears the house

Throttling the kids

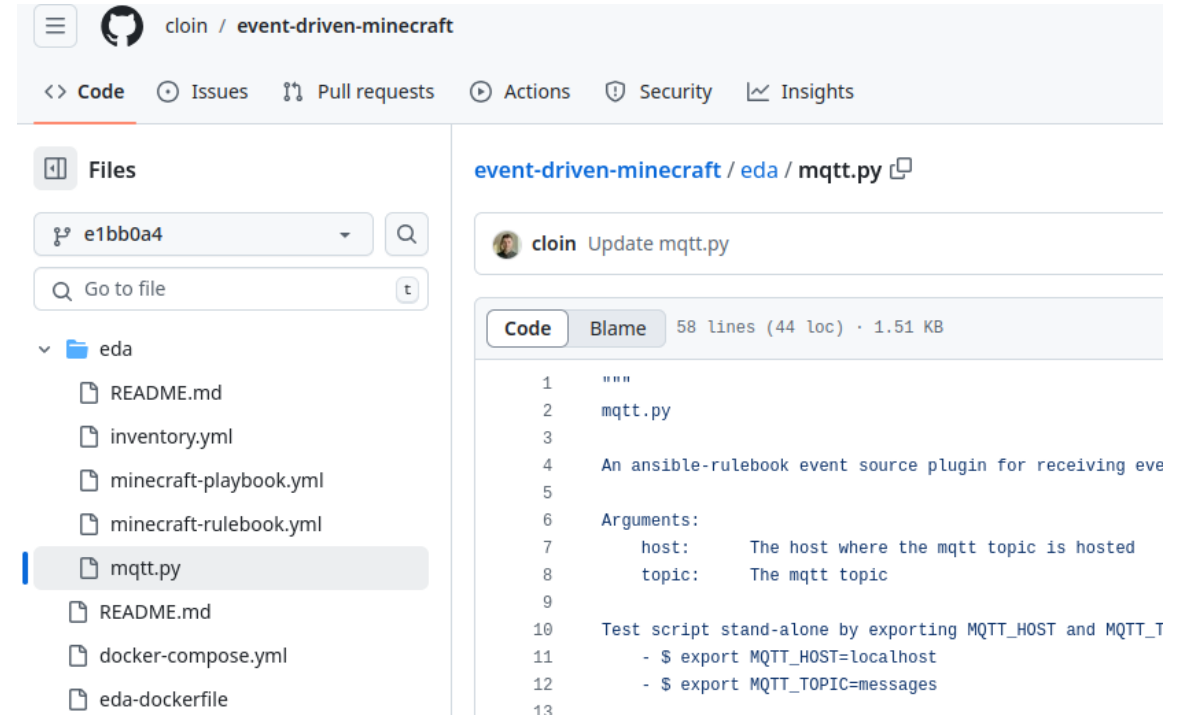
Imagine the following:

- Kids play Roblox/Fortnite/Among Us
- They become addicted and can't stop
- Let's limit Wi-Fi access EDA!



EDA and MQTT

- EDA uses event source plugins
- ...MQTT not supported by default
- Thank you, Colin McNaughton!
- Only needed a few changes



The screenshot shows the GitHub interface for the repository 'cloin / event-driven-minecraft'. The 'Files' tab is active, displaying a directory structure with files like 'README.md', 'inventory.yml', 'minecraft-playbook.yml', 'minecraft-rulebook.yml', 'mqtt.py', 'docker-compose.yml', and 'eda-dockerfile'. The 'mqtt.py' file is selected and its content is displayed in the main area. The file is a Python script for an Ansible event source plugin. It includes a docstring describing its purpose and arguments, and a test script at the bottom.

```
1  """
2  mqtt.py
3
4  An ansible-rulebook event source plugin for receiving eve
5
6  Arguments:
7      host:      The host where the mqtt topic is hosted
8      topic:     The mqtt topic
9
10 Test script stand-alone by exporting MQTT_HOST and MQTT_T
11 - $ export MQTT_HOST=localhost
12 - $ export MQTT_TOPIC=messages
13
```

Pipeline: Logs

```
Oct 28 17:19:49 dnsmasq[198080]: query[A] clientsettingscdn.roblox.com from 192.168.1.226
Oct 28 17:19:49 dnsmasq[198080]: forwarded clientsettingscdn.roblox.com to 208.67.222.222
Oct 28 17:19:49 dnsmasq[198080]: reply clientsettingscdn.roblox.com is <CNAME>
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.84
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.94
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.16
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.77
```

Pipeline: Logstash

```
input {  
  file {  
    path => ['/var/log/pihole.log']  
    mode => 'tail'  
    start_position => 'end'  
  }  
}  
  
filter {  
  mutate { gsub => ["message", "^.+?: ", ""] }  
  if ([message] !~ "^query") { drop {} }  
  dissect {  
    mapping => { message => "query[%{type}] %{name} from %{source}" }  
  }  
  prune { whitelist_names => ["^type$", "^name$", "^source$"] }  
  if ([source] =~ "^(127\.|172\.)") { drop {} }  
}  
  
output {  
  stdout { codec => 'json' }  
  mqtt {  
    host => "192.168.1.243"  
    port => 1883  
    topic => "dns"  
  }  
}
```

Pipeline: EDA

```
- name: mqtt events
hosts: localhost
sources:
  - runejuhl.weird_ansible.mqtt:
      host: '{{ mqtt_server }}'
      topic: 'dns'

rules:
  - name: Run job on DNS lookup
    condition:
      all:
        - event.payload.name == "clientsettingscdn.roblox.com"
          # match 21, 22, 23, 00, 01, 02, 03, 04, 05, 06, 07
        - event.meta.received_at is regex('T(2[123]|0[0-7]):')
    action:
      run_job_template:
        name: unifi_throttle
        organization: Default
        job_args:
          extra_vars:
            throttle_client: true
```

Pipeline: UniFi

```
name: |-
  Throttle wifi
hosts: localhost
gather_facts: false
tasks:

- name: Login
  ansible.builtin.uri:
    url: '{{ unifi_baseurl }}/api/login'
    method: POST
    headers:
      content-type: 'application/json'
    return_content: true
    body:
      {
        "username": "{{ unifi_username }}",
        "password": "{{ unifi_password }}",
        "remember": false,
        "strict": true
      }
    body_format: 'json'
  register: login

- name: Get devices
  ansible.builtin.uri:
    url: '{{ unifi_baseurl }}/v2/api/site/default/clients/active?includeTrafficUsage=true&includeUnifiDevices=true'
    headers:
      Cookie: |-
        {{ login.cookies_string }}
      X-Csrftoken: |-
        {{ login.cookies.csrf_token }}
    return_content: true
  register: devices

- ansible.builtin.set_fact:
  device: |-
    {{ devices.json|selectattr('ip', 'eq', ansible_edat.event.payload.source)|first }}

- name: Throttle device
  ansible.builtin.uri:
    url: '{{ unifi_baseurl }}/api/s/default/rest/user/{{ device.user_id }}'
    method: PUT
    headers:
      Cookie: |-
        {{ login.cookies_string }}
      X-Csrftoken: |-
        {{ login.cookies.csrf_token }}
    return_content: true
    body:
      usergroup_id: |-
        {% if throttle_client %}5ff6a0c246e0fb01138ed94f{% endif %}
    body_format: 'json'
  register: call_throttle

- ansible.builtin.debug:
  msg: |
    {{ call_throttle.json }}
  when: |-
    not call_throttle.json.meta.rc == "ok"
```

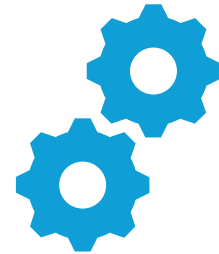

Loo roll detector



Common household
issue



Not fun being left out to
dry



A case for automation!

Technical specs

- NodeMCU dev board (ESP8266)
- TCS34725 RGB sensor
- Li-Ion 18650 battery holder



Pipeline: Server-Sent Events

- Event plugin based on MQTT plugin
- Added matching on event IDs

```
- name: http sse events
  hosts: localhost
  default_events_ttl: "2 seconds"
  match_multiple_rules: false
  sources:
    - http_sse:
        url: '{{ loo_roll_basename }}/events'
        event_ids:
          - sensor-tcs34725_color_temperature
          - sensor-tcs34725_illuminance
```



Home Assistant • now 



Loo roll empty, again!

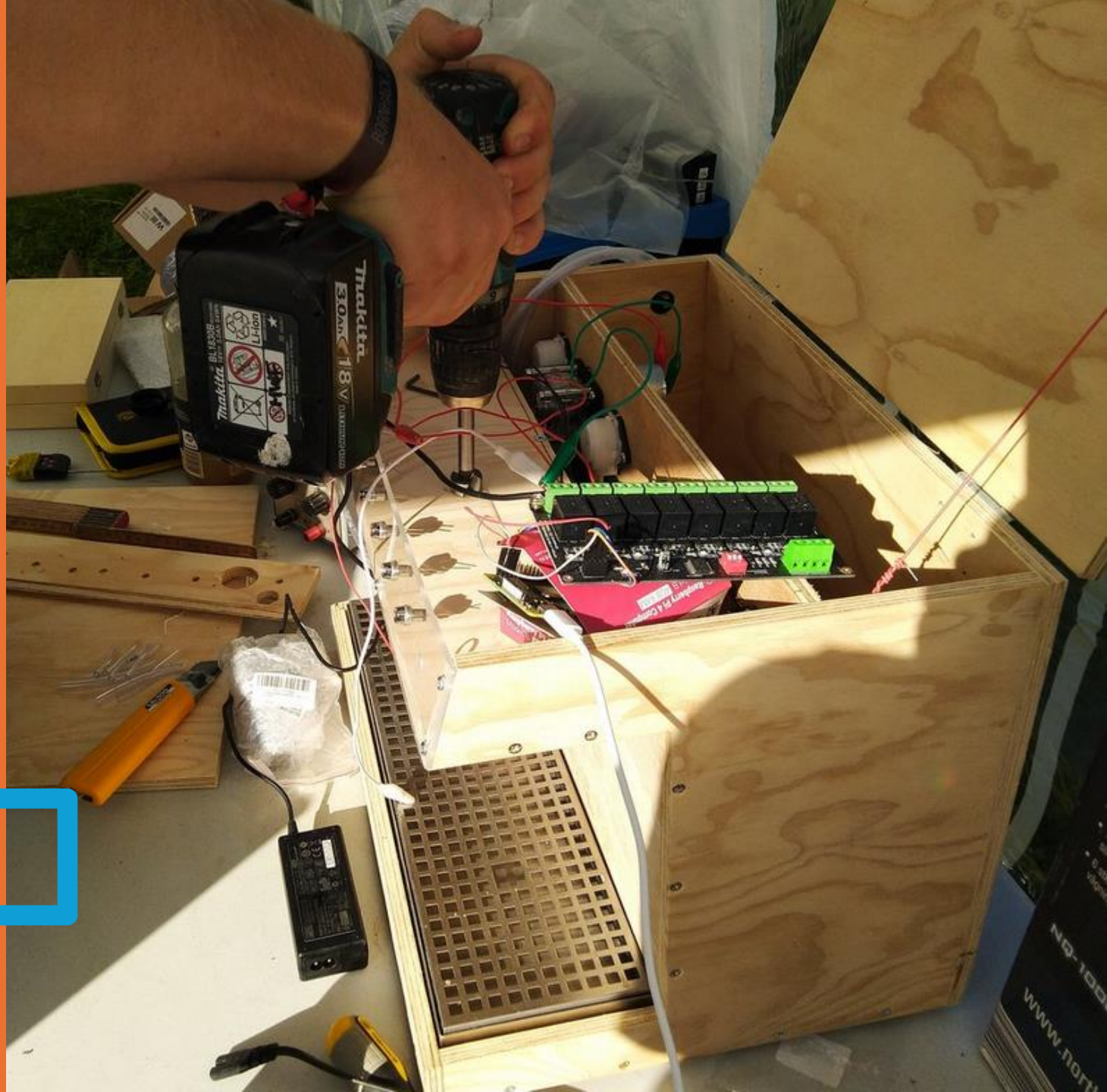
Whodunnit?

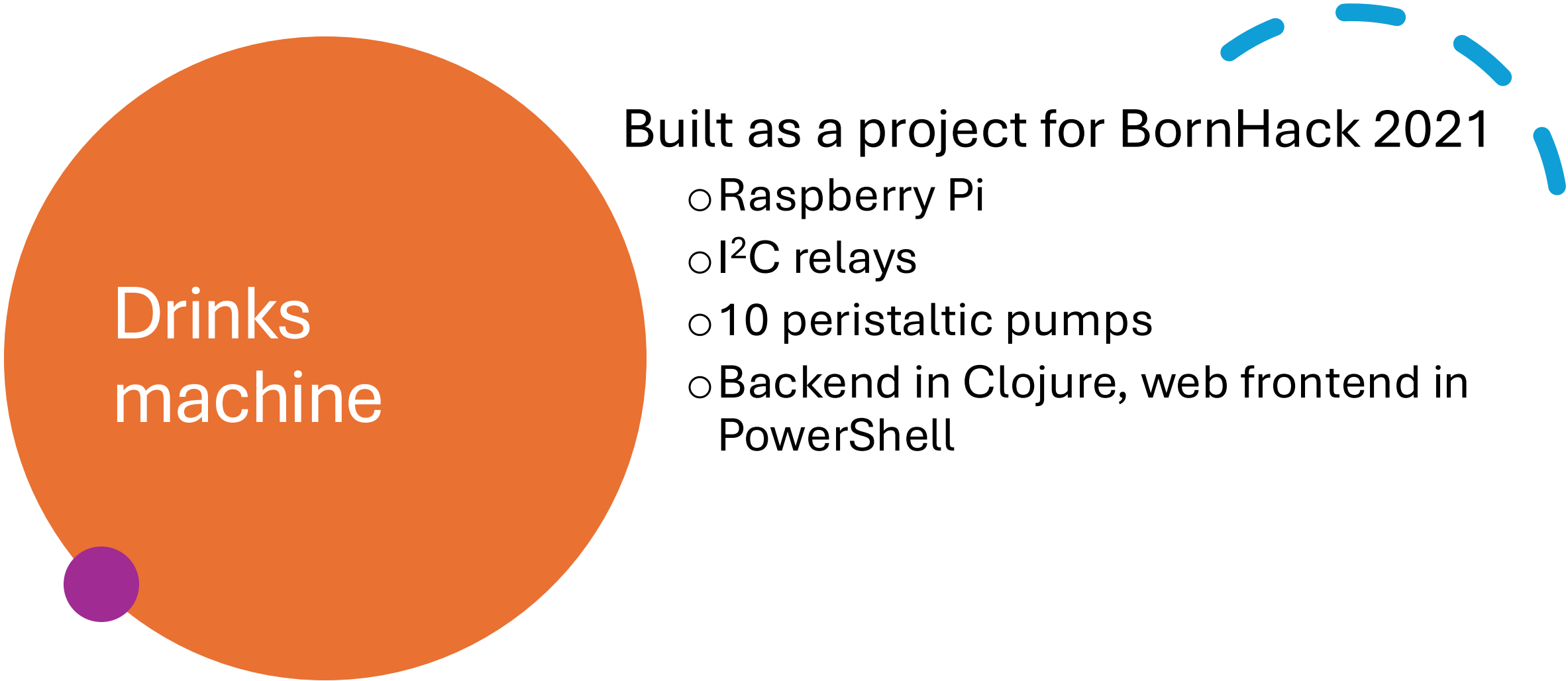


Improvements

- ESP32-based presence detector?
 - Bluetooth MACs aren't stable
- Facial recognition using Frigate and Double Take?
- ...maybe an accessible stock is easier.

Drinks machine





Drinks machine

Built as a project for BornHack 2021

- Raspberry Pi
- I²C relays
- 10 peristaltic pumps
- Backend in Clojure, web frontend in PowerShell

Status



Almost worked at the time



In storage ever since

...much to the
dismay of my kids
Until now!



Fixed the last bug (I hope)



Simple HTTP API (just POST, no body)

A private bartender

- Job templates allow us to limit access
 - Kids only allowed virgin cocktails
- Home Assistant REST API shows when I'm home
- EDA has url_check event plugin
 - Minor patch to allow authorization header

Templates ?

A job template is a definition and set of paramet

☐ ▾

▾

▾

Name ↑

- > ☐ Make adult drink
- > ☐ Make virgin drink

Templates > Make virgin drink

Prompt on Launch

1 Survey

2 Review

Which drink? *

Select option ▾

virgin-bloody-mary

virgin-pina-colada

orange-juice

exotic-fruits-juice



Troubles along the way

- AAP 2.5 containerized all-in-one mismatch with API URL in job template runner
- Decision Environment builds take *a lot* of disk space...
- Differences between ansible-rulebook and AAP EDA
- ...but I did what I set out to do!



That's it!

If the drinks machine worked during presentation, you're welcome to come up and have a (probably terrible) drink!